1. **Question on power system design and frequency regulation**

a) Assume that there are n identical generators, each with a capacity of 50 kW and availability of 0.98.

i) Explain the benefits of developing large-scale interconnected power systems for electricity supply.

[2 marks]

Power system reliability, security, economic efficiency, etc...

ii) The system design standard requires that the probability of meeting P kW peak demand is equal or greater than X. As a system design engineer, formulate a general optimization problem to determine the minimum number of the generators n_{min} that meet this standard (P is a parameter in the optimization).

[4 marks]

$$\min \ n \tag{1.1}$$
$$\text{s.t. } P \leq 50n \tag{1.2}$$
$$0.98^n + \sum_{k=1}^{n-1} (C_n^k 0.98^{n-k} * 0.02^k)\delta_k \geq \text{X} \tag{1.3}$$
$$n \in Z \tag{1.4}$$

where δ_k is defined as follows:

$$\delta_k = \begin{cases} 1, \text{ if } 50(n-k) \geq P \\ 0, \text{ else} \end{cases} \tag{1.5}$$

[4 marks]

iii) Assume that the coincidence factor of an infinite number of households is 0.333, the peak demand of each individual household is 4 kW, and the reliability criterion requires that the probability of meeting peak demand is equal to or greater than 99%. Based on the formulated optimization problem, find n_{min} to supply 90 households.

[4 marks]

$m = 90$, to meet 99% reliability requirement, the maximum peak demand that can be served is:

$$\left(0.333 + \frac{1-0.333}{\sqrt{m}}\right) m \times 4 = 145.19kW \tag{1.6}$$

[2 marks]

Solve the optimization in ii) with P = 145.19:

- $n = 3$, the probability of meeting 145.19 kW peak demand is $0.98^3 = 0.941192$, which is less than 0.99.
- $n = 4$, the probability of meeting 145.19 kW peak demand is $0.98^4 + 4 \times 0.98^3 \times 0.02 = 0.99766$, which is greater than 0.99.

Therefore , $n_{min} = 4$.

[2 marks]

b) Battery storage systems can provide frequency response much faster than thermal generators, therefore a new service called Dynamic Containment (DC) is designed in the UK. DC can effectively reduce the need for the regular frequency response (R) to maintain the frequency nadir as required, following the rule below:

$$\left(\frac{H}{f_0} - \text{DC} \times \frac{T_{\text{DC}}}{4 \times \Delta f_{\max}}\right) \times R \geq \frac{(P_L - \text{DC})^2 \times T_d}{4 \times \Delta f_{\max}} \tag{1.7}$$

Nomenclature:

H: System level of inertia (MW-s)
R: Amount of frequency response scheduled (MW)
DC: Amount of DCscheduled (MW)
P_{L}: Value of generation outage we want to secure against (MW)
f_0: Nominal frequency of the power grid (Hz)
$\Delta f_{\max}$: Maximum admissible frequency deviation (Hz)
T_{d}: Delivery time of frequency response (s)
T_{DC}: Delivery time of DC (s)

Assumptions:

- The amount of frequency response: $R = 2500$ MWs,
- The cost of frequency response: £5/MW,
- The largest generation unit in the UK: 1.8 GW,
- Other parameters: $f_0 = 50$ Hz, $\Delta f_{\max} = 1.0$ Hz, $T_d = 10$ s, $T_{\text{DC}} = 0.5$ s.

i) Please list three key index to describe the post-fault system frequency and discuss the potential reasons that the system may encounters large (in magnitude) deviations in those three index?

[3 marks]

- Frequency Nadir: H, DC, R, P_L
- Max RoCoF: H, P_L
- Steady-state frequency: R, DC, P_L

[1 mark each - 3 marks]

ii) Based on (1.1), calculate the minimum required inertia in the system to maintain the frequency nadir constraint. Do the same calculation when DC =200 MW is added to replace 200 MW frequency response R. Please comment the differences in the inertia response in those two cases

[3 marks]

- DC = 0 MW, H = 162.00 GWs
- DC = 200 MW, R = 2300 MWs, H = 140.38 GWs

[1 mark each - 2 marks]

DC can more effectively replace system inertia than frequency response

[1 mark]

iii) Based on the case with minimum required inertia and 0MW DC in (ii), assuming there is a bid of 150MW of DC, what is the maximum price (£/MW) the system operator should pay for it to achieve the minimum cost?

[4 marks]

As in ii), the system minumum required inertia is H = 162.00 GWs when DC = 0 MW. If there is a bid of 150 MW of DC, the frequency response can be reduced to $R' = 2112.9MW$, with $\Delta R = 387.1MW$.

The price of frequency response is £5/MW, therefore the maximum price of DC is

$$DC_{max} = \frac{387.1MW \times £5/MW}{150MW} = £12.9/MW \tag{1.8}$$

[4 marks]

2. **Question on per unit system**

a) What are the advantages to apply per-unit system in the analysis of power systems?

To avoid the confusion caused by:

- Multiple voltage levels

- Impedance of transformers depends on side

- Single-phase vs three phase

[3 marks]

b) Consider a three-phase high-voltage transmission circuit in Figure 2.2. A 60 MW, 0.8 lagging power-factor load is supplied from a 33 kV substation whose receiving bus is regulated at 30 kV.

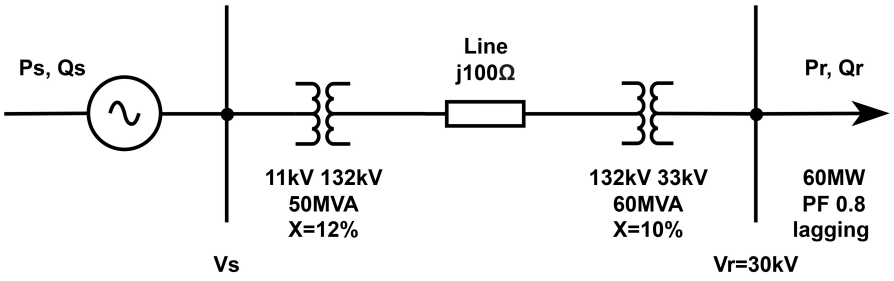


Figure 2.1 High-voltage transmission circuit with specified parameters.

i) Calculate the per-unit values of the quantities (three impedances) in the figure with properly selected bases for convenience, and sketch the equivalent circuit.

[5 marks]

Sketch:

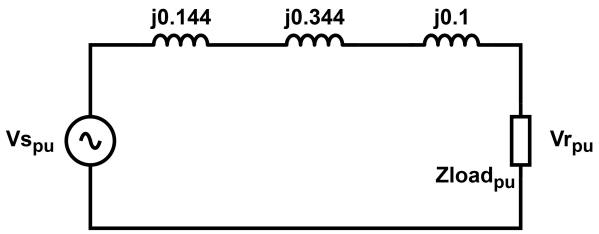


Figure 2.2 Equivalent circuit.

[1 mark]

Choose: $S_B = 60$ MW, $V_{B11} = 11$ kV, $V_{B132} = 132$ kV, $V_{B33} = 33$ kV.

[1 mark]

$$Z_{T1} = \text{j}0.12\frac{60}{50} = j0.144\,\text{p.u.} \qquad (2.1)$$

$$Z_{line} = \frac{\text{j}100}{V_{B132}^2/S_B} = j0.344\,\text{p.u.} \tag{2.2}$$

$$Z_{T2} = \text{j}0.1\,\text{p.u.} \tag{2.3}$$

[3 marks]

ii) Calculate the magnitude and phase angle of the voltage V_s at the sending end.

[5 marks]

Receiving end voltage in p.u. on 33 kV base:

$$V_r = \frac{30}{33} = 0.909\angle 0^\circ \text{ p.u.}$$

Load powers on 60 MVA base:

$$P_r = \frac{60}{60} = 1.00, \qquad S_r = \frac{P}{\text{PF}} = \frac{1}{0.8} = 1.25, \qquad Q_r = \sqrt{S^2 - P^2} = 0.75.$$

Load current:

$$I_r = \frac{P_r - jQ_r}{V_r} = \frac{1 - j0.75}{0.909} = 1.10 - j0.825 = 1.375\angle(-36.87^\circ)\text{ p.u.}$$

[3 marks]

Sending-end voltage:

$$V_s = V_r + jX_\Sigma I_r = 0.909 + j0.588(1.10 - j0.825) = 1.394 + j0.647\text{ p.u.}$$

$$|V_s| = 1.538,\ \angle V_s = 24.9^\circ\text{ (p.u.)}$$

[2 marks]

At 11 kV base (line-to-line): $|V_s| \approx 16.9$ kV.

iii) At the receiving end, when
(1) $P_r = 0$ and Q_r are set to 1 p.u. and 0.5 p.u. respectively,
(2) $Q_r = 0$ and P_r are set to 1 p.u. and 0.5 p.u. respectively,
calculate the magnitude and phase angle of the voltage V_s in these four scenarios again and state your findings.

[7 marks]

With $R \approx 0$, for real V_r the phasor relation simplifies to

$$V_s = V_r + \frac{X_\Sigma Q}{V_r} + j\frac{X_\Sigma P}{V_r}, \qquad (X_\Sigma = 0.588,\ V_r = 0.909).$$

[1 mark]

$$\text{(1a) } P = 0,\ Q = 1.0: \quad V_s = 0.909 + \frac{0.588 \cdot 1}{0.909} = 1.556\angle 0^\circ\text{ p.u.}$$

$$\text{(1b) } P = 0,\ Q = 0.5: \quad V_s = 0.909 + \frac{0.588 \cdot 0.5}{0.909} = 1.233\angle 0^\circ\text{ p.u.}$$

$$\text{(2a) } P = 1.0,\ Q = 0: \quad V_s = 0.909 + j\frac{0.588 \cdot 1}{0.909} = 0.909 + j0.647$$

$$|V_s| = 1.116,\ \angle V_s = 35.5^\circ$$

$$\text{(2b) } P = 0.5,\ Q = 0: \quad V_s = 0.909 + j\frac{0.588 \cdot 0.5}{0.909} = 0.909 + j0.324$$

$$|V_s| = 0.965,\ \angle V_s = 19.6^\circ$$

[4 marks]

Findings:
Active power flow is mainly driven by the phase-angle difference;
Reactive power flow is driven mainly by voltage-magnitude difference.

[2 marks]

3. **Question on power flow analysis and Gauss-Seidel method**

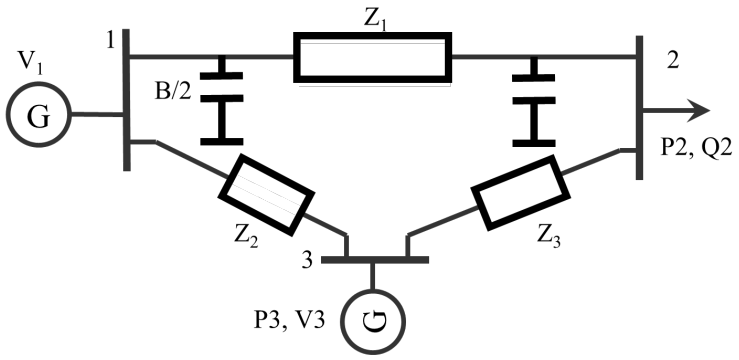


Figure 3.1 A three-bus power network

A three-bus power network is presented in Figure 3.1.

a) Express the bus admittance matrix for the system in terms of Z_1, Z_2, Z_3 and B.
[5 marks]

$$i_1 = j\frac{B}{2}V_1 + \frac{1}{Z_1}(V_1 - V_2) + \frac{1}{Z_2}(V_1 - V_3) = \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2}\right)V_1 - \frac{1}{Z_1}V_2 - \frac{1}{Z_2}V_3 \quad (3.1)$$

$$i_2 = j\frac{B}{2}V_2 + \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_3}(V_2 - V_3) = -\frac{1}{Z_1}V_1 + \left(j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_3}V_3 \quad (3.2)$$

$$i_3 = \frac{1}{Z_2}(V_3 - V_1) + \frac{1}{Z_3}(V_3 - V_2) = -\frac{1}{Z_2}V_1 - \frac{1}{Z_3}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_3}\right)V_3 \quad (3.3)$$

[3 marks]

$$Y = \begin{bmatrix} j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_2} & -\frac{1}{Z_1} & -\frac{1}{Z_2} \\ -\frac{1}{Z_1} & j\frac{B}{2} + \frac{1}{Z_1} + \frac{1}{Z_3} & -\frac{1}{Z_3} \\ -\frac{1}{Z_2} & -\frac{1}{Z_3} & \frac{1}{Z_2} + \frac{1}{Z_3} \end{bmatrix} \quad (3.4)$$

[2 mark]

b) Assuming the per unit values $|V_1| = 1, |V_3| = 1.05, P_3 = 1.03, P_2 = 0.96, Q_2 = 1, Z_1 = \mathrm{j}0.18, Z_2 = \mathrm{j}0.20, Z_3 = \mathrm{j}0.18, B = 0$, perform two iterations of the Gauss-Seidel load flow method.
[12 marks]

$$V_1 = 1\angle 0^\circ, \qquad V_2^0 = 1.00\angle 0^\circ, \qquad V_3^0 = 1.05\angle 0^\circ$$

[1 mark]

$$Y = \begin{pmatrix} -\mathrm{j}10.56 & \mathrm{j}5.56 & \mathrm{j}5.00 \\ \mathrm{j}5.56 & -\mathrm{j}11.11 & \mathrm{j}5.56 \\ \mathrm{j}5.00 & \mathrm{j}5.56 & -\mathrm{j}10.56 \end{pmatrix}.$$

[1 mark]

Iteration 1:

$$V_2^{(1)} = \frac{1}{Y_{22}}\left(\frac{S_2^*}{(V_2^0)^*} - Y_{21}V_1 - Y_{23}V_3^0\right) \tag{3.5}$$

$$= \frac{1}{-\mathrm{j}11.11}\left(\frac{-0.96+\mathrm{j}1}{1.00} - \mathrm{j}5.56 - \mathrm{j}5.56\times 1.05\right) \tag{3.6}$$

$$\approx 0.94\angle(-5.28^\circ). \tag{3.7}$$

[2 marks]

$$S_3^* = (V_3^0)^*\left(Y_{31}V_1 + Y_{33}V_3^0 + Y_{32}V_2^{(1)}\right) \tag{3.8}$$

$$= 1.05(\mathrm{j}5.00 - \mathrm{j}10.56\times 1.05 + \mathrm{j}5.56\times 0.94\angle(-5.28^\circ)) \tag{3.9}$$

$$\approx 0.50 - \mathrm{j}0.93, \qquad Q_3 = \mathrm{j}0.93. \tag{3.10}$$

[2 marks]

$$V_3^{(1)} = \frac{1}{Y_{33}}\left(\frac{P_3 - \mathrm{j}Q_3}{(V_3^0)^*} - Y_{31}V_1 - Y_{32}V_2^{(1)}\right) \tag{3.11}$$

$$= \frac{1}{-\mathrm{j}10.56}\left(\frac{1.03-\mathrm{j}0.93}{1.05} - \mathrm{j}5.00 - \mathrm{j}5.56\times 0.94\angle(-5.28^\circ)\right) \tag{3.12}$$

$$\approx 1.05\angle 2.59^\circ. \tag{3.13}$$

[2 marks]

Iteration 2:

$$V_2^{(2)} = \frac{1}{Y_{22}}\left(\frac{S_2^*}{(V_2^{(1)})^*} - Y_{21}V_1 - Y_{23}V_3^{(1)}\right) \tag{3.14}$$

$$= \frac{1}{-\mathrm{j}11.11}\left(\frac{-0.96+\mathrm{j}1}{0.94\angle 5.28^\circ} - \mathrm{j}5.56 - \mathrm{j}5.56\times 1.05\angle 2.59^\circ\right) \tag{3.15}$$

$$\approx 0.92\angle(-3.67^\circ). \tag{3.16}$$

[1 mark]

$$S_3^* = (V_3^{(1)})^*\left(Y_{31}V_1 + Y_{33}V_3^{(1)} + Y_{32}V_2^{(2)}\right) \tag{3.17}$$

$$= 1.05\angle(-2.59^\circ)(\mathrm{j}5.00 - \mathrm{j}10.56\times 1.05\angle 2.59^\circ + \mathrm{j}5.56\times 0.92\angle(-3.67^\circ)) \tag{3.18}$$

$$\approx 0.82 - \mathrm{j}1.05, \qquad Q_3 = \mathrm{j}1.05. \tag{3.19}$$

[1 mark]

$$V_3^{(2)} = \frac{1}{Y_{33}}\left(\frac{P_3 - \mathrm{j}Q_3}{(V_3^{(1)})^*} - Y_{31}V_1 - Y_{32}V_2^{(2)}\right) \tag{3.20}$$

$$= \frac{1}{-\mathrm{j}10.56}\left(\frac{1.03-\mathrm{j}1.05}{1.05\angle(-2.59^\circ)} - \mathrm{j}5.00 - \mathrm{j}5.56\times 0.92\angle(-3.67^\circ)\right) \tag{3.21}$$

$$\approx 1.05\angle 3.60^\circ. \tag{3.22}$$

[2 marks]

c) What factors could lead to a high voltage magnitude at Bus 3, and what mitigation measures would be taken in this case?

[3 marks]

More than enough reactive power is generated from G3, or the reactive power of the demand is too small. [1 mark]

Short term: e.g. Re-dispatch generation

Long term: e.g. Add sources of reactive power closer to load

[2 marks]

4. **Question on balanced fault analysis**

a) i)
- When two power systems are interconnected, a fault at any busbar is now supplied by all generators in both systems. Hence, the short-circuit level increases.
- Alternatively, the Thevenin equivalent impedance of the system decreases after interconnection, which increases the fault level.

[1 mark]

ii)
- Conventional thermal plants use synchronous generators with large electromagnetic coupling, capable of delivering high fault currents.
- Converter-interfaced (non-synchronous) generators, such as wind or solar inverters, have limited overcurrent capability (typically ≤ 1.5 p.u.), resulting in lower short-circuit levels.

[1 mark]

b)
- During a fault, the magnetic flux in a synchronous machine increases rapidly. Since the flux linking a closed winding cannot change instantaneously, the excess flux initially passes through high-reluctance paths.
- This results in low inductance and high current immediately after the fault.
- As the flux redistributes to lower-reluctance paths, the effective inductance increases and current reduces to steady-state.
- The sub-transient reactance represents this initial condition and captures the highest current immediately after the fault.

[1 mark each – 4 marks]

c) i) Base MVA:

$$S_{\text{base}} = 1000 \text{ MVA}$$

Equivalent reactances:

$$x_1 = \frac{S_{\text{base}}}{\text{SCL}_1} = \frac{1000}{5000} = 0.2 \text{ p.u.}, \quad x_2 = \frac{S_{\text{base}}}{\text{SCL}_2} = \frac{1000}{7500} = 0.133 \text{ p.u.}$$

[2 marks]

Equivalent reactance of two parallel lines:

$$x_{le} = \frac{x_l}{2} = 0.05 \text{ p.u.}$$

[1 mark]

Reactances after interconnection:

$$x_1' = \frac{x_1(x_{le}+x_2)}{x_1+x_{le}+x_2} = 0.0957 \text{ p.u.}, \quad x_2' = \frac{x_2(x_{le}+x_1)}{x_1+x_{le}+x_2} = 0.087 \text{ p.u.}$$

[2 marks]

Short-circuit levels:

$$\text{SCL}_1' = \frac{S_{\text{base}}}{x_1'} = 10{,}455 \text{ MVA}, \quad \text{SCL}_2' = \frac{S_{\text{base}}}{x_2'} = 11{,}500 \text{ MVA}$$

[2 marks]

ii) Using Δ–Y conversion:

$$x_a = x_b = \frac{x_l(x_l/2)}{x_l + x_l/2 + x_l/2} = \frac{x_l}{4} = 0.025 \text{ p.u.}$$

$$x_c = \frac{(x_l/2)^2}{x_l + x_l/2 + x_l/2} = \frac{x_l}{8} = 0.0125 \text{ p.u.}$$

[3 marks]

Equivalent reactance at fault point:

$$X_{\text{eq}} = x_c + (x_a + x_1) \parallel (x_b + x_2) = 0.1054 \text{ p.u.}$$

[1 mark]

Fault current:

$$I_f = \frac{1}{X_{\text{eq}}} = 9.4845 \text{ p.u.}$$

[1 mark]

Base current:

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{\text{base}}} = \frac{1000}{\sqrt{3} \times 275} = 2.1 \text{ kA}$$

[1 mark]

Fault current in kA:

$$I_f = 9.4845 \times 2.1 = 19.91 \text{ kA}$$

[1 mark]

5. **Question on unbalanced fault analysis**

a) Given:

$$I_A = -(I_B + I_C) = 1.51\angle -138.64^\circ \text{ p.u.}$$

[1 mark]

Zero-sequence component:

$$I_{A0} = I_{B0} = I_{C0} = \frac{1}{3}(I_A + I_B + I_C) = 0$$

[1 mark]

Positive-sequence component:

$$I_{A1} = \frac{1}{3}(I_A + \alpha I_B + \alpha^2 I_C) = 0.814\angle -136.17^\circ \text{ p.u.}$$

$$I_{B1} = \alpha^2 I_{A1} = 0.814\angle -103.83^\circ, \quad I_{C1} = \alpha I_{A1} = 0.814\angle 16.17^\circ$$

[2 marks]

Negative-sequence component:

$$I_{A2} = \frac{1}{3}(I_A + \alpha^2 I_B + \alpha I_C) = 0.693\angle -141.53^\circ \text{ p.u.}$$

$$I_{B2} = \alpha I_{A2} = 0.693\angle -21.53^\circ, \quad I_{C2} = \alpha^2 I_{A2} = 0.693\angle 98.47^\circ$$

[2 marks]

b) i) Equivalent sequence reactances:

$$X_{1eqN} = (X_{1L} + X_{1T} + X_{1G}) \parallel (X_{1T} + X_{1G}) = 0.1505 \text{ p.u.}$$

$$X_{2eqN} = (X_{2L} + X_{2T} + X_{2G}) \parallel (X_{2T} + X_{2G}) = 0.1505 \text{ p.u.}$$

[1 mark]

Zero-sequence network:

$$X_{0eqN} = (X_{0L} + X_{0T}) \parallel (X_{0T}) = 0.1031 \text{ p.u.}$$

[1 mark]

Fault current:

$$I_f = \frac{3}{jX_{1eqN} + jX_{2eqN} + jX_{0eqN}} = -j7.424 \text{ p.u.}$$

[1 mark]

Base current:

$$I_{\text{base}} = \frac{S_{\text{base}}}{\sqrt{3}V_{\text{base}}} = \frac{500}{\sqrt{3} \times 133} = 2.171 \text{ kA}$$

[1 mark]

Fault current magnitude:

$$|I_f| = 7.424 \times 2.171 = 16.11 \text{ kA}$$

[1 mark]

ii)

$$V_B = \alpha^2 \left(1 - jX_{1eqN}\frac{I_f}{3}\right) + \alpha \left(-jX_{2eqN}\frac{I_f}{3}\right) + \left(-jX_{0eqN}\frac{I_f}{3}\right)$$

[3 marks]

$$= 0.947\angle 125.93^\circ \text{ p.u.}$$

[1 mark]

$$|V_B| = 0.947 \times 133 = 125.93 \text{ kV}$$

[1 mark]

iii)

$$X_{1eqL} = X_{1L} + X_{1T} + X_{1G} = 0.34 \text{ p.u.}$$
$$X_{2eqL} = X_{2L} + X_{2T} + X_{2G} = 0.34 \text{ p.u.}$$

[1 mark]

$$X_{0eqL} = X_{0L} + X_{0T} = 0.35 \text{ p.u.}$$

[1 mark]

$$I_f = \frac{3}{jX_{1eqL} + jX_{2eqL} + jX_{0eqL}} = -j2.913 \text{ p.u.}$$

[1 mark]

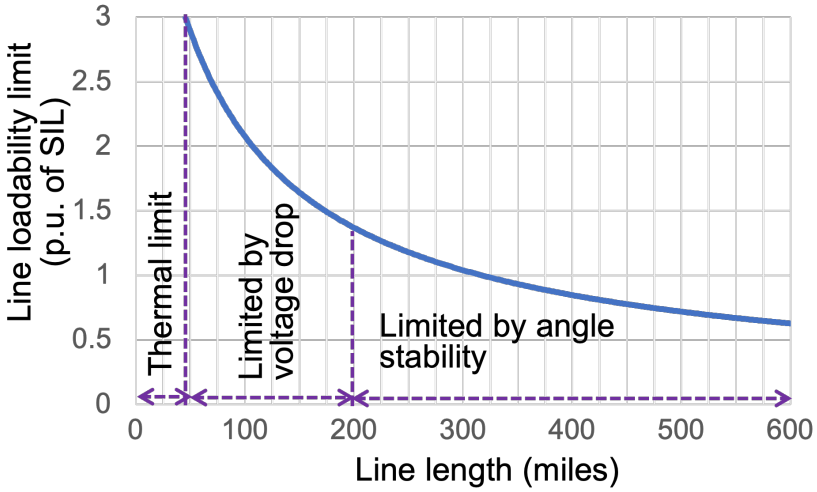
$$|I_f| = 2.913 \times 2.171 = 6.32 \text{ kA}$$

[1 mark]

6. **Question on stability analysis**

a)
- For lines up to 100 km: limited by thermal capacity.
- 100–300 km: limited by voltage drop or voltage stability.
- Beyond 320 km: limited by angle stability.

[2 marks]



[2 marks]

b) Incremental change in electrical power output

$$\Delta P_e = \left(\frac{\partial P_e}{\partial \delta}\right)_0 \Delta\delta$$

[1 mark]

Consider incremental change from the swing equation

$$\frac{2H}{\omega_s}\frac{d^2(\Delta\delta)}{dt^2} = -\Delta P_e$$

[1 mark]

Combining the above

$$\frac{2H}{\omega_s}\frac{d^2(\Delta\delta)}{dt^2} + \left(\frac{\partial P_e}{\partial \delta}\right)_0 \Delta\delta = 0$$

[1 mark]

$$\omega_n = \sqrt{\frac{(\partial P_e/\partial\delta)_0 \omega_s}{2H}}$$

[2 marks]

c) i)

$$P = 0.9 \text{ p.u.}, \quad \theta = \sin^{-1}\left(\frac{0.9\times 0.3}{1}\right) = 15.66^\circ$$

[1 mark]

$$I = \frac{1\angle\theta - 1}{j0.3}$$

[1 mark]

$$E = 1\angle\theta + j0.2\left(\frac{1\angle\theta - 1}{j0.3}\right) = 1.04 \text{ p.u.}$$

[1 mark]

ii)

$$\sin\delta_0 = \frac{0.9(0.2+0.3)}{1.04} \Rightarrow \delta_0 = 25.64^\circ$$

[2 marks]

$$\omega_n = \sqrt{\frac{(1.04/0.5)\cos 25.64^\circ \times 50}{8}} = 1.37 \text{ Hz}$$

[2 marks]

iii) Using equal area criterion

$$\int_{\delta_0}^{\delta_{cr}} (P-0)d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{\max}\sin\delta - P)d\delta$$

[2 marks]

$$\delta_{cr} = \cos^{-1}\left[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0\right] = 85.95^\circ$$

[2 marks]